Seat No.

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2 **Year**: 2024

Subject: 031021112: Electrical Measuring Instruments **Section**: 5-6

Date: 11 March 2025 **Time**: 8:00-10:00

Name________________________________ID: ____________________Class_________

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections with 48 questions. There will be 16 pages (including this page) and they are worth 55 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

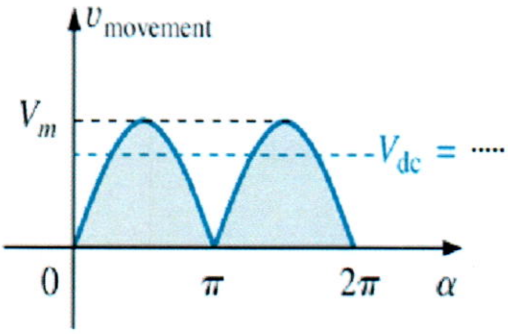
Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:		Q1-12	Q11-14	Q15-31
CLO 1	Describe the basic knowledge about principles and operation of various type of electrical instruments and measurements i.e. Oscilloscope, Signal generator, Wattmeter, Power Factor meter and Watt-hour meter.	√		
CLO 2	Explain and design the various types of Bridge including Inductance and capacitance bridge.		√	
CLO 3	Explain the concept of resistance measurement by using V-A method, meter testing, meter calibrating, and extending measurement scales.			√

Section 1: Answer the questions

Q1) Find the V_{Dc} (V_{Ave}) of **Full-wave rectified waveform** by using **Integration.**

(3 points)



Integration Formulas:

$$\int 1dx = x + c$$
$$\int \sin x\, dx = -\cos x + c$$
$$\int \cos x\, dx = \sin x + c$$

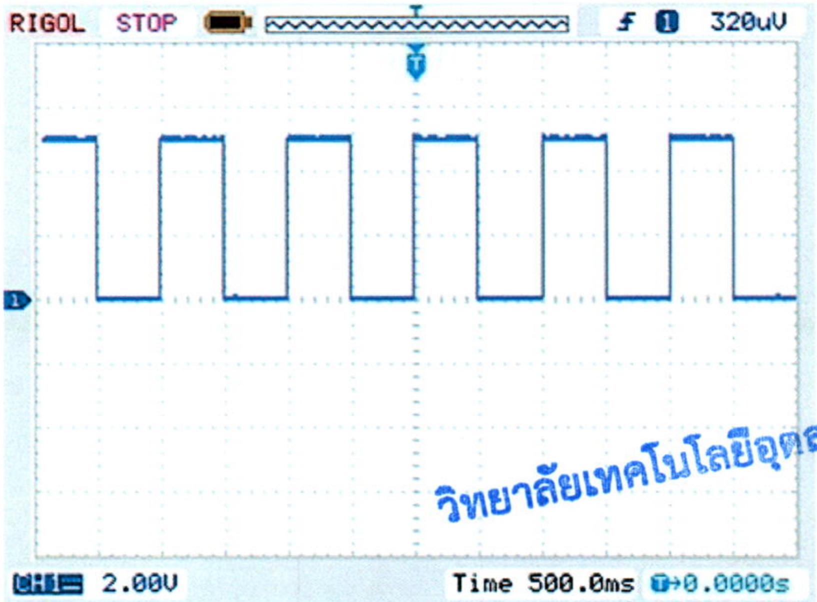
Q2) Draw the **basic function generator** block diagram. (2 points)

Q3) Draw and explain the simplified block diagram of both ***analog*** and ***digital*** **oscilloscopes**. (4 points)

Q4) Show how to calibrate an **analog Voltmeter**. (2 points)

Section 2: Multiple Choice Question

Q1) Determine the **Pulse Width (PW)**, **Space Width (SW)** of the signal waveform where TIME/DIV = 500 ms and VOLTS/DIV = 2 V.



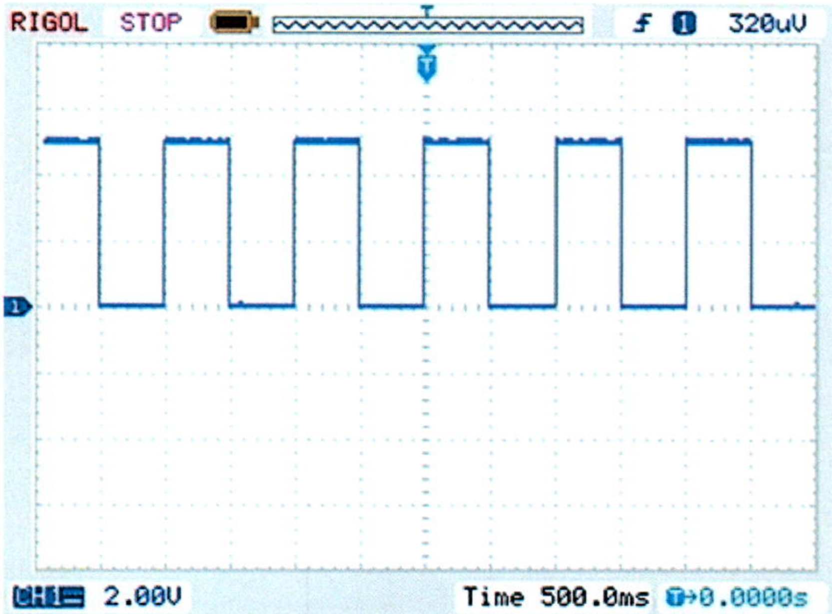
a) PW = 500 ms SW = 500 ms

b) PW = 500 ms SW = 1000 ms

c) PW = 1000 ms SW = 500 ms

d) PW = 1000 ms SW = 1000 ms

Q2) Determine the **Pulse Amplitude (PA)** and **Frequency (f)** of the signal waveform where TIME/DIV = 500 ms and VOLTS/DIV = 2 V.



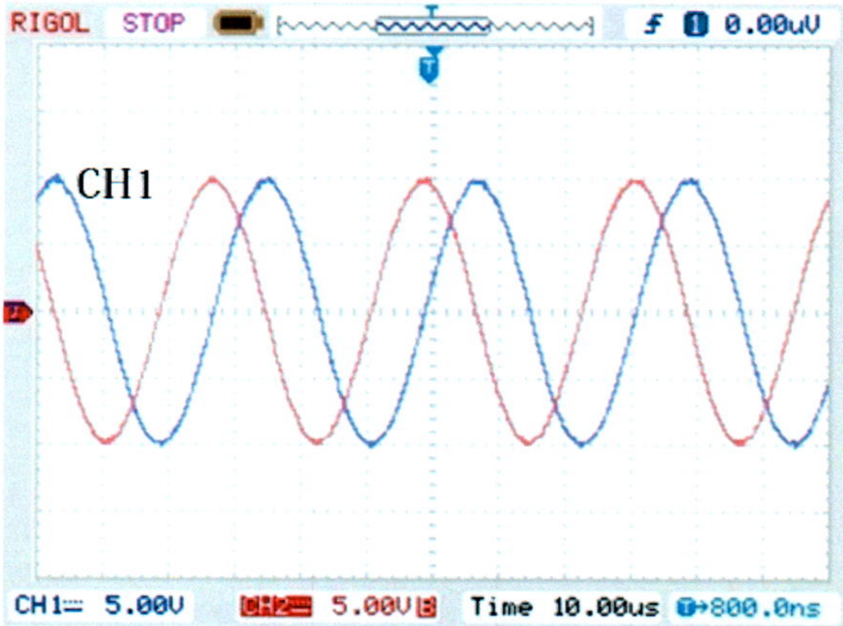
a) PA = 5 V f = 1 Hz

b) PA = 5 V f = 2 Hz

c) PA = 2.5 V f = 500 Hz

d) PA = 2.5 V f = 1000 Hz

Q3) Determine the **amplitude (Vrms) and Volt Peak (VP) of the signal from CH1** where TIME/DIV = 10 μs and VOLTS/DIV = 5 V.



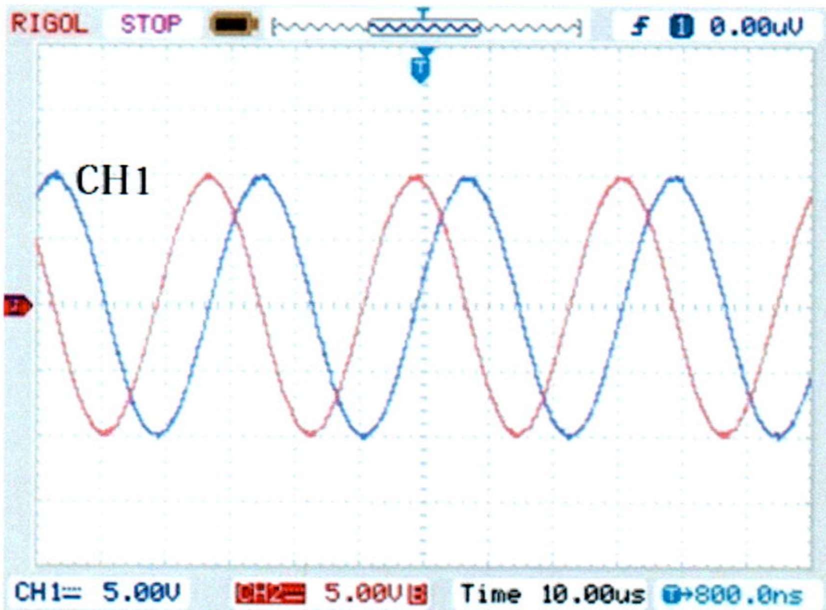
a) Vrms = 7.07 V VP = 10 V

b) Vrms = 7.07 V VP = 20 V

c) Vrms = 5.77 V VP = 10 V

d) Vrms = 6.36 V VP = 20 V

Q4) Determine the <u>**phase difference**</u> between the two waveforms where TIME/DIV = 10 μs and VOLTS/DIV = 5 V.



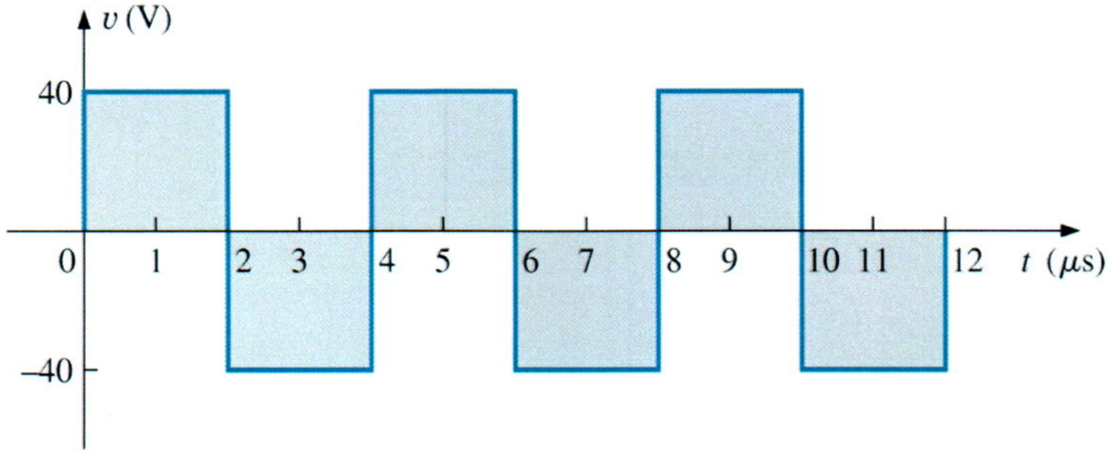
a) Phase difference (ϕ) = 90°

b) Phase difference (ϕ) = 180°

c) Phase difference (ϕ) = 60°

d) Phase difference (ϕ) = 120°

Q5) Determine the Frequency of the waveform of the signal waveform.



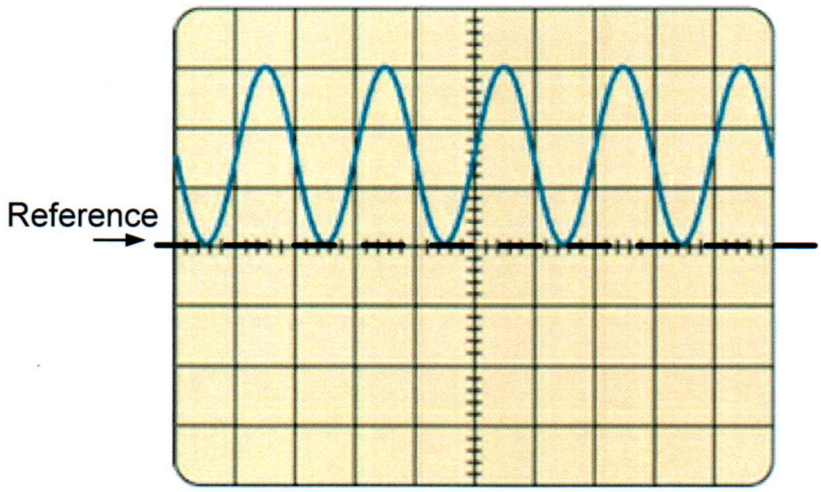
a) f= 250 kHz

b) f= 0.25 kHz

c) f= 250 Hz

d) f= 250 MHz

Q6) From the oscilloscope display, choose the possible **input coupling selector setting** of the oscilloscope.



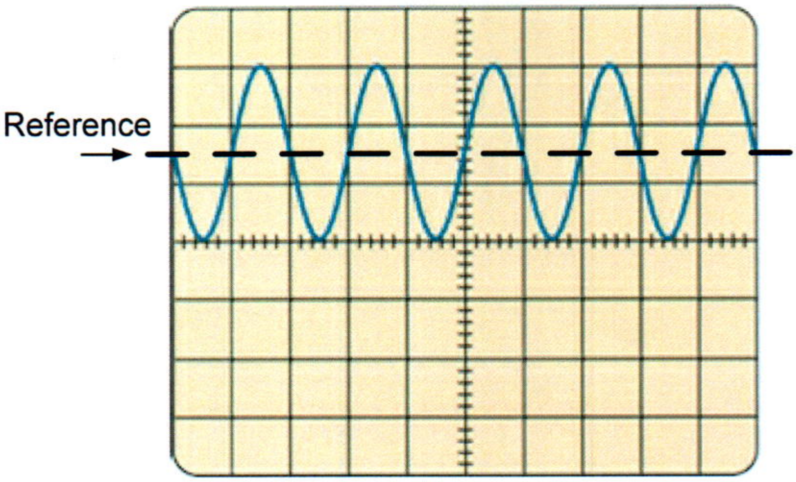
a) DC

b) AC

c) DC and AC

d) GND

Q7) From the oscilloscope display, choose the possible **input coupling selector setting** of the oscilloscope.



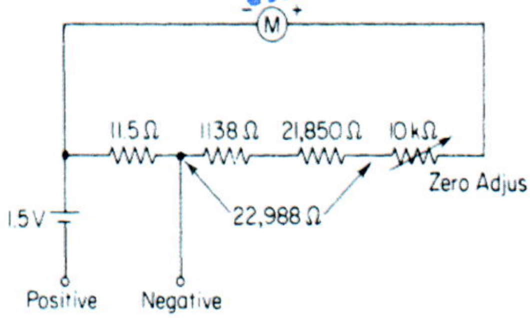
a) DC

b) AC

c) DC and AC

d) GND

Q8) From the schematic diagrams of the Simpson Model 260 multimeter, list the name of circuits.



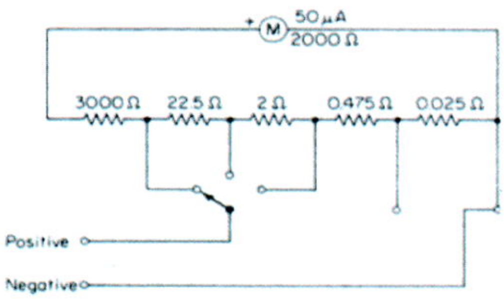
a) Voltmeter

b) Ammeter

c) Ohmmeter

d) Wattmeter

Q9) From the schematic diagrams of the Simpson Model 260 multimeter, list the name of circuits.



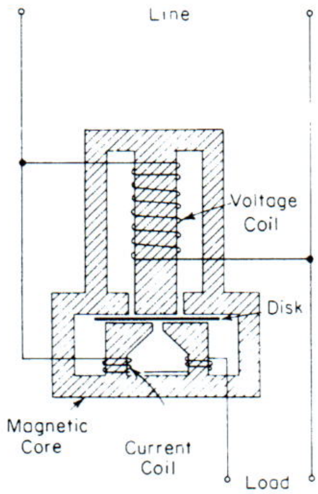
a) Voltmeter

b) Ammeter

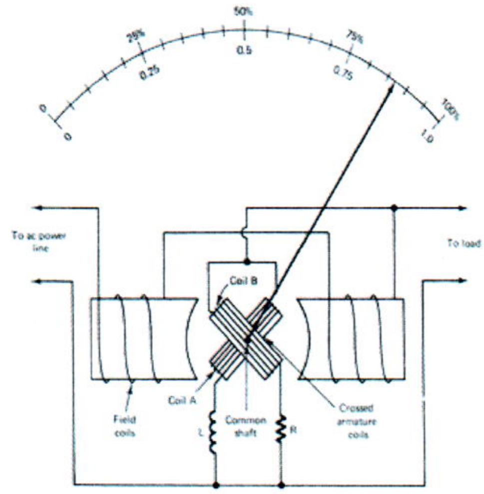
c) Ohmmeter

d) Wattmeter

Q10) List the name of the instrument (1) and (2) shown in the schematic form.



(1)



(2)

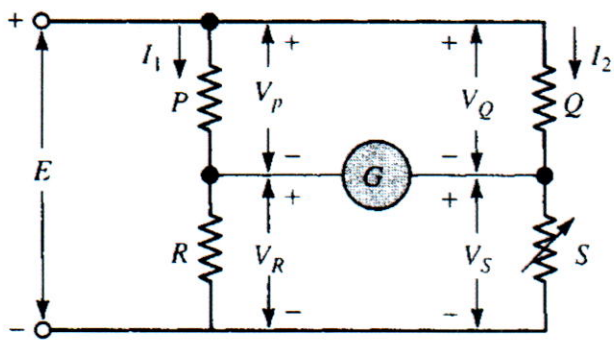
a) (1) Watt-hour Meter and (2) Power Factor Meter

b) (1) Power Factor Meter and (2) Watt-hour Meter

c) (1) Watt-hour Meter and (2) Frequency Meter

d) (1) Clamp-on Meter and (2) Power Factor Meter

Q11) Calculate the value of R, a Wheatstone bridge as shown in the Figure has P = 5 kohm, Q = 2 kohm, S = 10 kohm and galvanometer null is obtained (Bridge balance) .



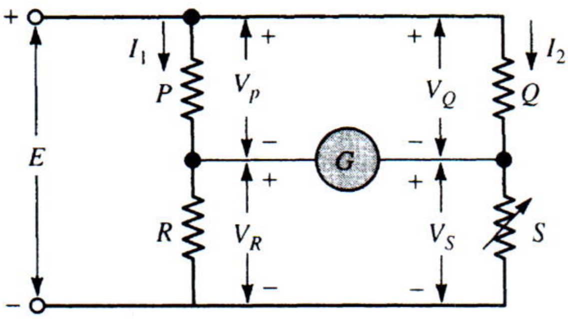
a) 1 kohm

b) 4 kohm

c) 25 kohm

d) 100 kohm

Q12) A Wheatstone bridge as in Figure has P = 5 kohm, Q =2 kohm, and galvanometer null is obtained. Determine the resistance measurement range (**R**) for the bridge if **S** is adjustable from 1 kohm to 15 kohm.



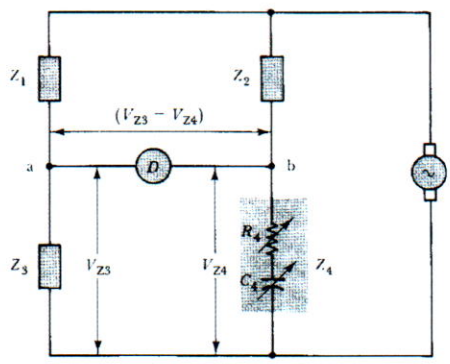
(a) 0 ohm - 25 kohm

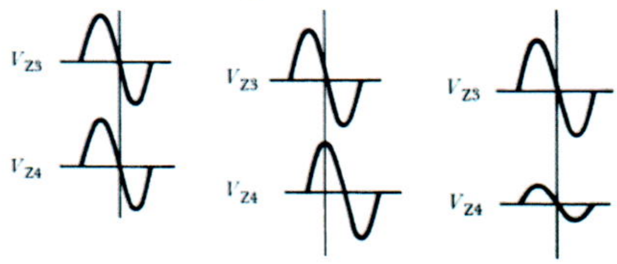
(b) 2.5 kohm - 37.5 kohm

(c) 2.5 kohm - 25 kohm

(d) 25 kohm – 37.5 kohm

Q13) From the figures, identify each of the case of AC bridge **balanced** or **unbalanced**.





1) .... 2)......................... 3).........................

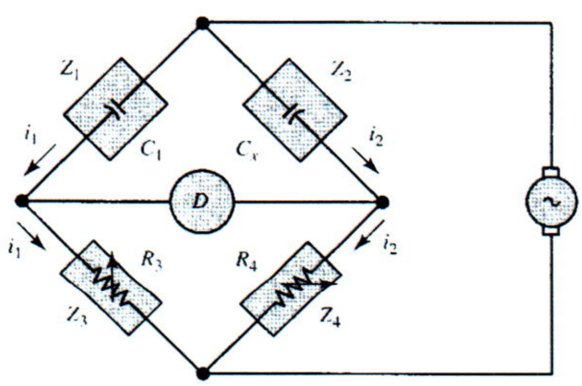
a) (1) Balanced (2) Unbalanced (3) Unbalanced

b) (1) Balanced (2) Unbalanced (3) Balanced

c) (1) Unbalanced (2) Balanced (3) Unbalanced

d) (1) Unbalanced (2) Balanced (3) Balanced

Q14) A simple capacitance bridge in Figure, uses a standard capacitor (C1) = 0.1 uF (Micro Farad), and R3/R4 can be set to any ratio between 200:1 and 1:200. Calculate the range of measurements of unknown capacitance Cx.



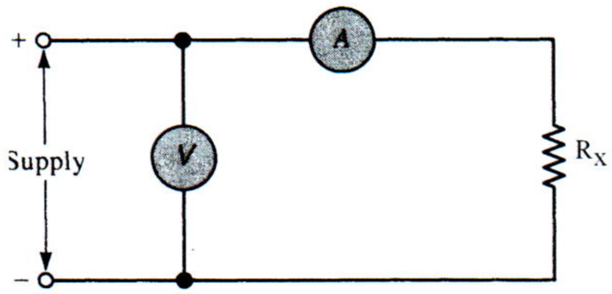
a) 0.5 nF - 20 uF

b) 0.5 uF -20 uF

c) 20 nF - 5 uF

d) 20 uF – 5 mF

Q15) A resistance is measured by the ammeter and voltmeter circuit illustrated in Figure. The measured current is 0.5 A, and the voltmeter indication is 500 V. The ammeter has a resistance of R_A = 10 ohm , and the voltmeter on a 1000 V range has a sensitivity of 10 kohm/V. Calculate the value of R_x by taking the effect of internal resistance (R_A and R_V) into account.



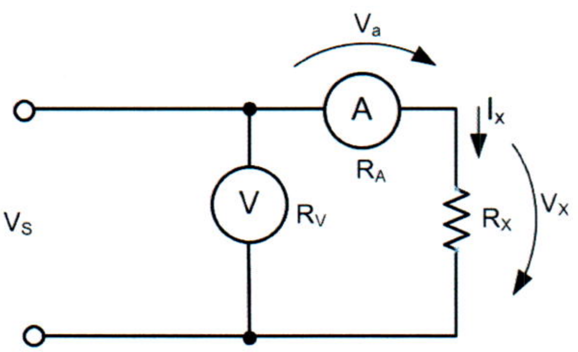
a) 990 ohm

b) 1000 ohm

c) 1010 ohm

d) 5 Mohm

Q16) From the circuit diagrams, **identify the equation of the error** if **Rx'** is the resistance from the measurement, **Rx** is the real value of Load.



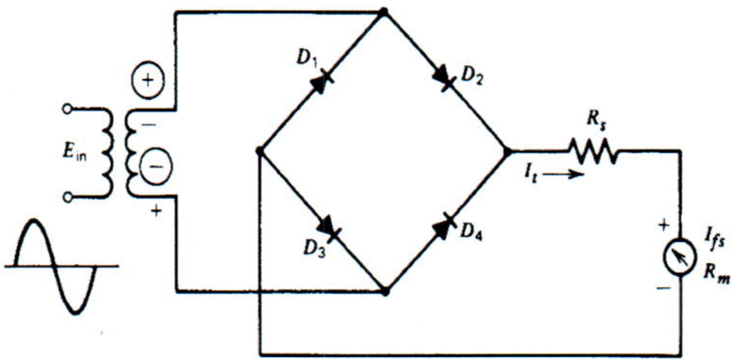
a) Error= RA/RX

b) Error= RX/RV

c) Error= RA/RV

d) Error= RV/RA

Q17) From the figure, **compute** the value of the multiplier resistor (R_s) for a 30-Vrms ac range on the voltmeter, where a PMMC instrument with FSD (I_{fs})= 500 uA (Micro Amm) and meter coil resistance (R_m) is 600 ohm.



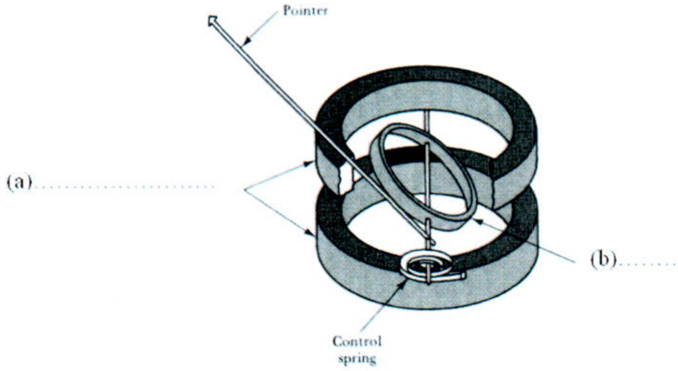
a) 53.4 kohm

b) 26.4 kohm

c) 41.8 kohm

d) 35.56 kohm

Q18) Identify each part of the electrodynamic instrument.



a) (a) Field Coil (b) Moving Coil

b) (a) Moving Coil (b) Field Coil

c) (a) Voltage Coil (b) Current Coil

d) (a) Current Coil (b) Voltage Coil

Q19) From the Power factor meter in Figure, determine Apparent Power (S), while Active Power (P) = 50 W.

a) 42.5 VA

b) 58.82 VA

c) 49.15 VA

d) 50.85 VA

Q20) Form the Power factor meter in Figure, determine 1) phase different between Active Power (P) and Apparent Power (S), while Active Power (P) = 50W.

a) 31.78 Degree

b) 58.21 Degree

c) 40.36 Degree

d) 49.15 Degree

Q21) Form the Power factor meter in Figure, determine Reactive Power (Q), while Active Power (P) = 50 W.

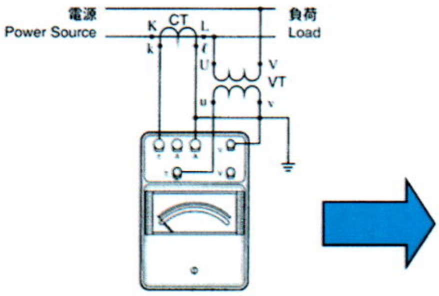
a) 30.98 VAR

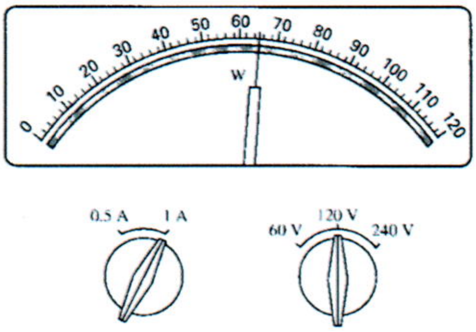
b) 42.5 VAR

c) 26.33 VAR

d) 58.82 VAR

Q22) For the wattmeter, CT, and VT in Figure, determine the measured power (watt) where: **VT ratio = 24000/120 V, CT ratio = 200/1 A,** respectively.





	Voltage Range	
	120 V	240 V
Current Range	Multiplier Constant	
1 A	1	2
5 A	5	10

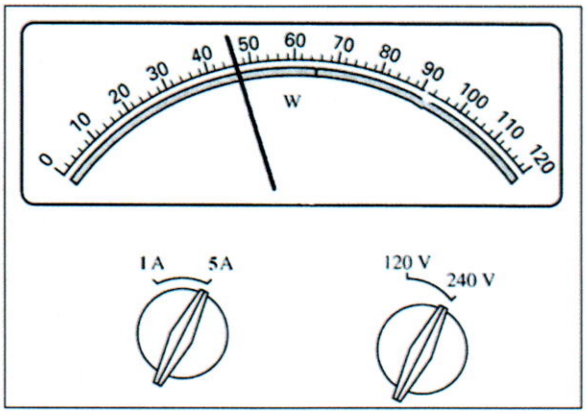
a) 2,600 kW

b) 5,200 kW

c) 13,000 W

d) 26,000 W

Q23) For the wattmeter in Figure, determine the measured power (watt).



	Voltage Range	
	120 V	240 V
Current Range	Multiplier Constant	
1 A	1	2
5 A	5	10

a) 460 W

b) 46 W

c) 230 W

d) 92 W

Q24) What is the main function of Current transformers (CT)?

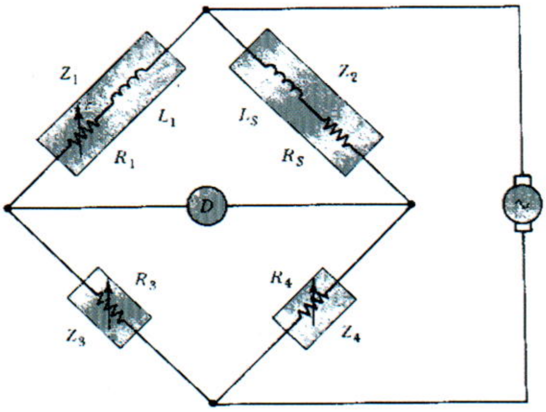
a) Extend the range of the current measuring instrument.

b) Extend the range of the voltage measuring instrument.

c) Isolate the measuring instrument between current and voltage.

d) All a. b. c.

Q25) An inductor that is marked as (LS) 100 mH is to be measured on an inductance comparison bridge. The bridge uses a 200 mH standard inductor for L1, and a 5 ohm standard resistor for R4. If the coil resistance of the (LS) 100 mH inductor is measured as (RS) 300 ohm, determine the resistances of R1 and R3 at which balance is likely to occur.



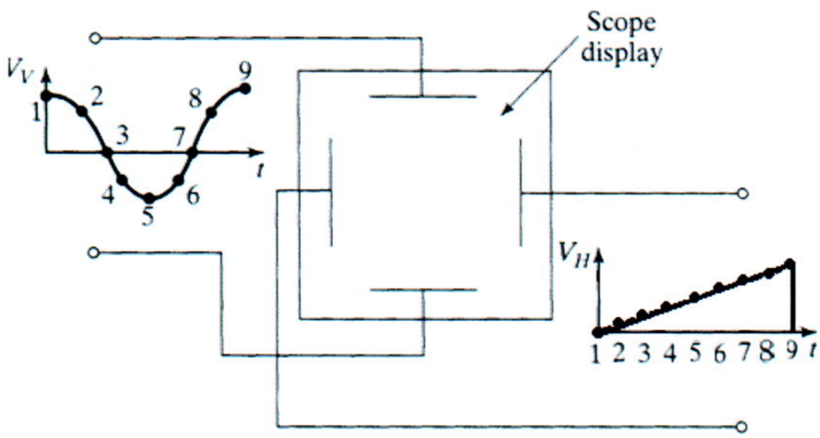
a) R1=600 ohm, R3=10 kohm

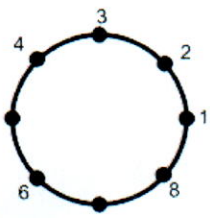
b) R1=10 kohm, R3=600 kohm

c) R1=150 ohm, R3=2.5 kohm

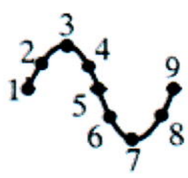
d) R1=2.5 kohm, R3= 150 ohm

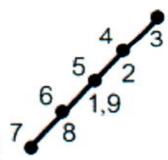
Q26) Which is the scope display that ***Lissajous Patterns*** are generated.



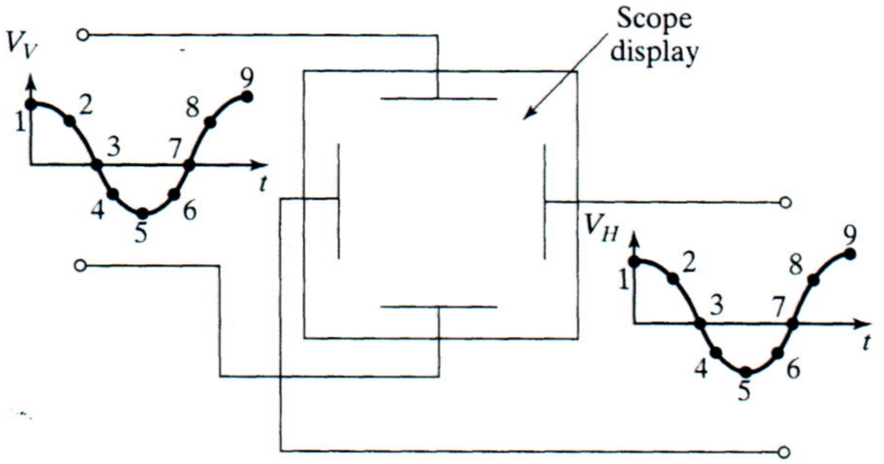
a) 

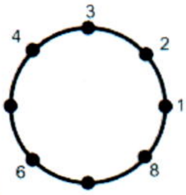
b) 

c) 

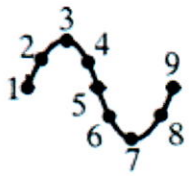
d) 

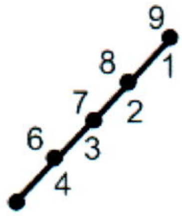
Q27) Which is the scope display that ***Lissajous Patterns*** are generated.



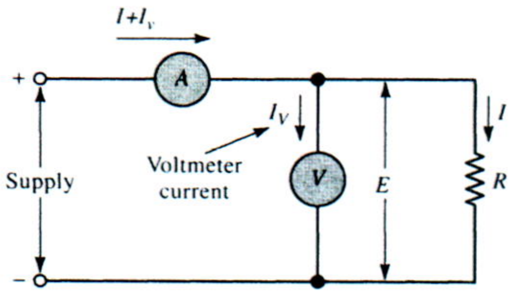
a) 

b) 

c) 

d) 

Q28) From the circuit diagrams, <u>identify</u> each of the Voltmeter and Ammeter methods and <u>specific</u> its operation of resistance measurement.



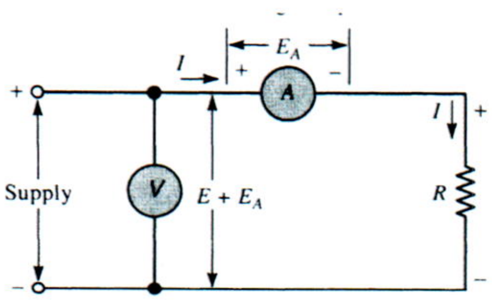
a) Correct Volt for Low Resistance (R)

b) Correct Volt for High Resistance (R)

c) Correct Amp for Low Resistance (R)

d) Correct Amp for High Resistance (R)

Q29) From the circuit diagrams, <u>identify</u> each of the Voltmeter and Ammeter methods and <u>specific</u> its operation of resistance measurement.



a) Correct Volt for Low Resistance (R)

b) Correct Volt for High Resistance (R)

c) Correct Amp for Low Resistance (R)

d) Correct Amp for High Resistance (R)

Q30) What is the function of (1) **Vertical Amplifier** and (2) **Timebase Generator** of an analog oscilloscope?

a) (1) Drive the vertical input and (2) Generates a sawtooth signal

b) (1) Drive the sawtooth signal and (2) Generates a sinewave signal

c) (1) Convert the analog signal to digital form and (2) Generate the stored waveform data.

d) (1) Generate the stored waveform data and (2) Convert the analog signal to digital form.

Section 3: True False Questions

Q1) The analog-to-digital converter (ADC) converts the analog signal into digital form at each pulse of the sample clock.

Q2) The display processor converts the stored waveform data into the sawtooth signal.

Q3) The function of the electron gun to produce the electron beam.

Q4) The characteristics of electronic devices can be displayed by using Y-T mode.

Q5) Oscilloscope probes are calibrated by connecting to a sine wave.

Q6) The accuracy of the standard instrument should be at least four times better than the instrument to be calibrated.

Q7) DC Voltmeter Calibration is also include a current-limiting resistor.

Q8) Current coil and Voltage coil of Wattmeter calibration is connected to the same dc power supply.

Q9) A function generator produces sine, square, and triangular waveform outputs.

Q10) The power factor is the cosine of the phase angle between voltage and current.

Q11) The electrodynamic movement is extensively used to measure AC power only.

Q12) The Balanced condition of Wheatstone bridge are not only the voltages equal in amplitude, they are also equal in phase.

Q13) Megohmmeter circuit utilize the correct volt concept.

Q14) A major advantage of the electrodynamic instrument is that it is not polarized.

Assoc. Prof. Dr. Pichet Sriyanyong

Answer Sheet

Final Examination of Semester 2 **Year:** 2024

Subject: 031021112 Electrical Measuring Instruments **Section:** 5-6

Name________________________________ID ____________________ Seat No.________

Section 2: Multiple Choice Questions

Students: *Fill circles completely with ink or pencil.

*Erase all stray marks completely.

	A	B	C	D	E
1	○	○	○	○	○
2	○	○	○	○	○
3	○	○	○	○	○
4	○	○	○	○	○
5	○	○	○	○	○
6	○	○	○	○	○
7	○	○	○	○	○
8	○	○	○	○	○
9	○	○	○	○	○
10	○	○	○	○	○

■	A	B	C	D	E
11	○	○	○	○	○
12	○	○	○	○	○
13	○	○	○	○	○
14	○	○	○	○	○
15	○	○	○	○	○
16	○	○	○	○	○
17	○	○	○	○	○
18	○	○	○	○	○
19	○	○	○	○	○
20	○	○	○	○	○

■	A	B	C	D	E
21	○	○	○	○	○
22	○	○	○	○	○
23	○	○	○	○	○
24	○	○	○	○	○
25	○	○	○	○	○
26	○	○	○	○	○
27	○	○	○	○	○
28	○	○	○	○	○
29	○	○	○	○	○
30	○	○	○	○	○

Section 3: True False Questions

Students: *Fill circles completely with ink or pencil

*Erase all stray marks completely.

	T	F
1	○	○
2	○	○
3	○	○
4	○	○
5	○	○
6	○	○
7	○	○

	T	F
8	○	○
9	○	○
10	○	○
11	○	○
12	○	○
13	○	○
14	○	○